

Ministry of Science and Technology
Department of Technical and Vocational Education
Promotion Examination for A.G.T.I. Year I

Date: 26.10.06

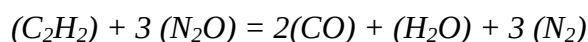
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Met-04011 Metallurgical Thermodynamics I

Answer any five.

1 (a) The combustion of acetylene fuel with nitrous oxide as oxidant at 25°C, is widely performed in flame emission spectrometry. Calculate the maximum temperature attained if the best mixture corresponds to the reaction:



Assume that water is undissociated.

$$\Delta H_{298}^{\circ} (C_2H_2) = 54.23 \text{ kcal/mole}, \quad \Delta H_{298}^{\circ} (N_2O) = 19.70 \text{ kcal/mole},$$

$$\Delta H_{298}^{\circ} (CO) = -26.42 \text{ kcal/mole}, \quad \Delta H_{298}^{\circ} (H_2O) = -57.80 \text{ kcal/mole},$$

$$C_p \text{ for } (CO) = 6.8 + 1.0 \times 10^{-3}T - 0.11 \times 10^{-5}T^2 \text{ cal/deg/mole}$$

$$C_p \text{ for } (H_2O) = 7.17 + 2.56 \times 10^{-3}T - 0.08 \times 10^{-5}T^2 \text{ cal/deg/mole}$$

$$C_p \text{ for } (N_2) = 6.5 + 1.0 \times 10^{-3}T \text{ cal/deg/mole}$$

Ans: $(C_2H_2) + 3 (N_2O) = 2(CO) + (H_2O) + 3 (N_2)$

$$\begin{aligned} \Delta H_{298}^{\circ} (\text{Reaction}) &= 2 \Delta H_{CO}^{\circ} + \Delta H_{H_2O}^{\circ} + 3 \Delta H_{N_2}^{\circ} - (\Delta H_{C_2H_2}^{\circ} + \Delta H_{N_2O}^{\circ}) \\ &= (2 \times -26.42) + (-57.80) + (0) - (54.23) - (3 \times 19.70) \\ &= -223.97 \text{ Kcal} \end{aligned}$$

$$\text{Heat evolved} = \int_{298}^{T_m} \sum C_{p_{\text{product}}} dT$$

$$223970 = \int_{298}^{T_m} (2C_{p_{CO}} + C_{p_{H_2O}} + 3C_{p_{N_2}}) dT$$

$$\begin{aligned} &= \int_{298}^{T_m} \left[(13.6 + 2.0 \times 10^{-3}T - 0.22 \times 10^{-5}T^2) + \right. \\ &\quad \left. (7.17 + 2.56 \times 10^{-3}T + 0.08 \times 10^{-5}T^2) + (19.5 + 3.0 \times 10^{-3}T) \right] dT \\ &= \int_{298}^{T_m} (40.27 + 7.56 \times 10^{-3}T - 0.14 \times 10^{-5}T^2) dT \\ &= \left[40.27T + 3.78 \times 10^{-3}T^2 + 0.14 \times 10^{-5}T^{-1} \right]_{298}^{T_m} \end{aligned}$$

$$223970 = 40.27 (T_m - 298) + 3.78 \times 10^{-3} (T_m^2 - 298^2) + 0.14 \times 10^5 \left(\frac{1}{T_m} - \frac{1}{298} \right)$$

Neglecting T_m^{-1}

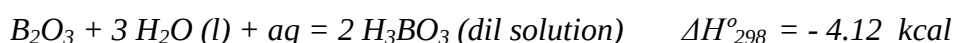
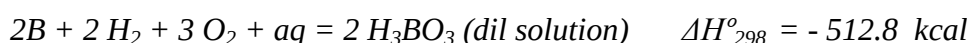
$$223970 = 3.78 \times 10^{-3} T_m^2 + 40.27 T_m - 12383$$

$$3.78 \times 10^{-3} T_m^2 + 40.27 T_m - 236353 = 0$$

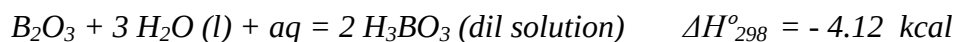
$$T_m = 4207 \text{ K}$$

$$\text{Maximum temperature} = 4207 - 273 = 3934^\circ\text{C}$$

The enthalpy changes for the following reactions are as follows:



I (b) Calculate the standard heat of formation of B_2O_3 in terms of per mole of B_2O_3 , and per gram of B_2O_3 . Atomic weight of B and O are 10.82 and 16.0 respectively.



$$\Delta H_{298}^\circ = -302.49 \text{ kcal}$$

$$\text{Relative molecular mass of } B_2O_3 = 2 \times 10.82 + 16 \times 3$$

$$= 69.64 \text{ gm/mole}$$

$$\text{Standard ht of formation of } B_2O_3 = -302.49 \text{ Kcal/mole}$$

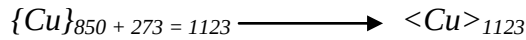
$$= -\frac{302.49}{69.64} \text{ Kcal/gm}$$

$$= -4.3436 \text{ Kcal/gm}$$

2 (a) Calculate the entropy change of the system and surroundings for the isothermal freezing of one mole of super cooled liquid silver at 850°C, when the surroundings are also at the same temperature.

Heat of fusion at the melting point of 1234°K is 2690 cal/mole.

C_p for solid silver is $5.09 + 2.04 \times 10^{-3}T$ cal/deg/mole and liquid copper is 7.3 cal/deg/mole.



$$\Delta S_{\text{sys}} = ?, \Delta S_{\text{syr}} = ?$$

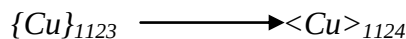
$$\Delta F_{\text{fus}} = 2690 \text{ cal/mole}, T_{\text{fus}} = 1234K$$

$$C_{p(s)} = 5.09 + 2.04 \times 10^{-3}T, \quad C_{p(l)} = 7.3 \text{ cal/deg/mole}$$

$$\text{Ans: } \{Cu\}_{1123} \longrightarrow \langle Cu \rangle_{1123} \quad \Delta S_1 = \int_{1123}^{1124} \frac{C_{p_l} dT}{T}$$

$$\{Cu\}_{1124} \longrightarrow \langle Cu \rangle_{1124} \quad \Delta S_2 = \frac{\Delta H_t}{T_t}$$

$$\langle Cu \rangle_{1124} \longrightarrow \langle Cu \rangle_{1123} \quad \Delta S_3 = \int_{1124}^{1123} C_{p_s} dT$$



$$\Delta S_{\text{sys}} = \Delta S_1 + \Delta S_2 + \Delta S_3 = \int_{1123}^{1234} \frac{7.3}{T} dT + \frac{-2690}{1234} + \int_{1234}^{1123} \frac{(5.09 + 2.04 \times 10^{-3}T)}{T} dT$$

$$= 7.3 \ln \frac{1234}{1123} - \frac{2690}{1234} + 5.09 \ln \frac{1123}{1234} + 1.02 \times 10^{-3} (1123 - 1234)$$

$$= 0.688 - 2.1799 - 0.479769 - 0.11322$$

$$= -2.084789$$

$$-\Delta H_{\text{sur}} = \Delta H_{\text{sys}} = \int_{1123}^{1234} 7.3 dT - 2690 + \int_{1234}^{1123} (5.09 + 2.04 \times 10^{-3}T) dT$$

$$= 7.3(1234 - 1123) - 2690 + 5.09(1123 - 1234) + 1.02 \times 10^{-3} (1123^2 - 1234^2)$$

$$= 810.3 - 2690 - 564.99 - 266.859$$

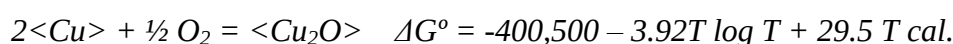
$$= -2711.549 \text{ cal}$$

$$\Delta S_{\text{sur}} = \frac{(-\Delta H_{\text{sys}})}{T} = \frac{+2711.549}{1234} = +2.19736 \text{ cal/deg}$$

$$\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{sur}} = +2.19736 - 2.084789$$

$$= 0.1125769 \text{ cal/deg}$$

2 (b) Calculate the standard enthalpy and entropy change at 298 K for the following reaction



Ans: $2\text{Cu} + \frac{1}{2} \text{O}_2 = \text{Cu}_2\text{O}$

$$\Delta G^\circ = -400,500 - 3.92T \log T + 29.5 T \text{ cal}$$

$$\frac{\Delta G^\circ}{T} = -\frac{400,500}{T} - 3.92 \log T + 29.5$$

$$\frac{\partial(\Delta G^\circ / T)}{\partial T} = \frac{400,500}{T^2} - 1.702 \frac{1}{T}$$

$$-\frac{\Delta H^\circ}{T^2} = \frac{400,500}{T^2} - \frac{1.702}{T}$$

$$\Delta H^\circ = -400,500 + 1.702T$$

$$= -400,500 + 1.702 \times 298$$

$$= -400,500 + 507$$

$$= -399,993 \text{ cal}$$

$$= -399.993 \text{ Kcal}$$

$$\frac{\partial \Delta G^\circ}{\partial T} = -1.702 - 1.702 \ln T + 29.5$$

$$\Delta S^\circ = 27.798 - 1.702 \ln T$$

$$\Delta S_{298}^\circ = 27.798 - 1.702 \ln 298$$

$$= -18.10 \text{ cal/deg}$$

3.(a) The melting point of gallium is 303K at 1atm. The densities of solid and liquid gallium are 5.885 g/cc and 6.08 g/cc. the heat of fusion of gallium is 18.5 cal/g. Calculate the change in melting point of gallium for an increase of pressure of 1 atm.

Ans:
$$\frac{dP}{dT} = \frac{\Delta H_{\text{fus}}}{T(V_l - V_s)}$$

Let atomic wt. Of gallium = x

$$\frac{dP}{dT} = \frac{18.5x \times 41.293}{303 \left(\frac{x}{\rho_l} - \frac{x}{\rho_s} \right)}$$

$$\rho_l = 6.08 \text{ g/cc}, \quad \rho_s = 5.885 \text{ g/cc},$$

$$1 \text{ cal} = 41.293 \text{ cc atm}$$

$$dP = 1 \text{ atm}, \quad dT = ?$$

$$\begin{aligned} \frac{dP}{dT} &= \frac{18.5x \times 41.293}{303 \left(\frac{x}{6.08} - \frac{x}{5.885} \right)} \\ &= -462.7 \text{ atm/deg} \end{aligned}$$

$$\frac{dT}{dP} = -2.16 \times 10^{-3} \text{ deg/atm}$$

$$dT = -2.16 \times 10^{-3} \text{ deg at } dP = 1 \text{ atm}$$

$\therefore$ melting point of gallium will be decreased by $2.16 \times 10^{-3} \text{ deg}$

3.(b) In an investigation for the thermodynamic properties of a manganese, the following heat contents were determined:

$$H_{700}^\circ - H_{298}^\circ = 2895 \text{ cal/g-atom}$$

$$H_{1000}^\circ - H_{298}^\circ = 2895 \text{ cal/g-atom}$$

Find applications suitable equation for $(H_T^\circ - H_{298}^\circ)$ and also for C_p as function of temperature in the form $(a + bT)$. Assume that no structural transformation takes place in the given temperature range.

Ans:
$$H_{700}^\circ - H_{298}^\circ = 2895 \text{ cal/g-atom}$$

$$H_{1000}^{\circ} - H_{298}^{\circ} = 5450 \text{ cal/g-atom}$$

$$C_p = a + bT$$

$$H_{700}^{\circ} - H_{298}^{\circ} = \int_{298}^{700} (a + bT) dT$$

$$2895 = \left[aT + \frac{bT^2}{2} \right]_{298}^{700}$$

$$2895 = a(700 - 298) + \frac{b}{2}(700^2 - 298^2)$$

$$2895 = 402a + 200598b$$

$$a + 499b = 7.2 \quad \dots\dots\dots(1)$$

$$H_{1000}^{\circ} - H_{298}^{\circ} = a(1000 - 298) + \frac{b}{2}(1000^2 - 298^2)$$

$$5450 = 702a + 455598b$$

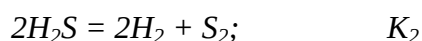
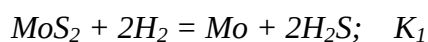
$$a + 649b = 7.7635 \quad \dots\dots\dots(2)$$

$$(1) \times (2) \Rightarrow b = 3.7566 \times 10^{-3}$$

$$a = 5.3254$$

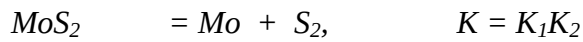
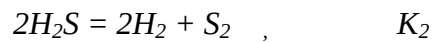
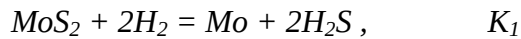
$$\begin{aligned} \therefore H_T^{\circ} - H_{298}^{\circ} &= a(T - 298) + \frac{b}{2}(T^2 - 298^2) \\ &= 5.3254(T - 298) + \frac{3.7566 \times 10^{-3}}{2}(T^2 - 298^2) \\ &= 5.3254T + 1.8783 \times 10^{-3}T^2 - 1753.76 \\ &= 1.8783 \times 10^{-3}T^2 + 5.3254T - 1753.76 \end{aligned}$$

4.(a) Calculate the vacuum needed for obtaining molybdenum metal according to the reaction



At temperature 800°C K_1 is 7.3×10^{-6} and K_2 is 2.2×10^{-4} and at temperature 1000°C K_1 is 1.439×10^{-3} and K_2 is 5.6×10^{-3} .

Solution



$$K = \frac{a_{\text{Mo}} P_{\text{S}_2}}{a_{\text{MoS}_2}} = P_{\text{S}_2}$$

$$\begin{aligned} \text{at } 800^\circ\text{C} \rightarrow \quad K_1 &= 7.3 \times 10^{-6}, K_2 = 2.2 \times 10^{-4} \\ K &= K_1 K_2 = 7.3 \times 10^{-6} \times 2.2 \times 10^{-4} \\ K &= P_{\text{S}_2} = 1.606 \times 10^{-9} \end{aligned}$$

$$\begin{aligned} \text{at } 1000^\circ\text{C} \rightarrow \quad K_1 &= 1.439 \times 10^{-3}, K_2 = 5.6 \times 10^{-3} \\ K &= K_1 K_2 = 1.439 \times 10^{-3} \times 5.6 \times 10^{-3} \\ K &= P_{\text{S}_2} = 8.0584 \times 10^{-6} \end{aligned}$$

4.(b) Calculate the mean of vaporization of palladium and vapor pressure at 1568°C from the following vapor pressure data : $7.516 \times 10^{-7} \text{ mmHg}$ at temperature 1314°C and $1.164 \times 10^{-6} \text{ mmHg}$ at 1351°C .

Ans: $\Delta H_v = ?$, $T = 1568 + 273 = 1841\text{K} \rightarrow P = ?$

$$T_1 = 1314 + 273 = 1587\text{K}, \quad p_1 = 7.516 \times 10^{-7} \text{ mmHg}$$

$$T_2 = 1351 + 273 = 1624\text{K}, \quad p_2 = 1.164 \times 10^{-6} \text{ mmHg}$$

$$\ln \frac{P_2}{P_1} = \frac{-\Delta H_v}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln \frac{1.164 \times 10^{-6}}{7.516 \times 10^{-7}} = \frac{-\Delta H_v}{1.987} \left(\frac{1}{1624} - \frac{1}{1587} \right)$$

$$\Delta H_v = 145871 \text{ cal}$$

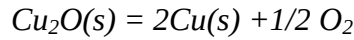
$$\begin{aligned} \ln \frac{P}{7.516 \times 10^{-7}} &= \frac{-145871}{1.987} \left(\frac{1}{1841} - \frac{1}{1587} \right) \\ &= 6.73198 \end{aligned}$$

$$P = 6.304 \times 10^{-4}$$

5.(a) Determine the lowest temperature at which copper oxide (Cu_2O) can dissociate in a vacuum of 10^{-5}mmHg .

Ans: $\text{Cu}_2\text{O(s)} = 2\text{Cu(s)} + 1/2 \text{O}_2 \quad \Delta G^\circ = 40500 + 3.92T \log T - 29.5T$

Solution



$$\Delta G^\circ = 40500 + 3.92T \log T - 29.5T$$

$$\Delta G^\circ = -RT \ln \frac{P_{\text{O}_2}^{1/2} \cdot a_{\text{Cu}}^2}{a_{\text{Cu}_2\text{O}}} = -4.575T \log P_{\text{O}_2}^{1/2}$$

In a vacuum of 10^{-5}mmHg , taking P_{O_2} in air as 0.2 atm

$$P_{\text{O}_2} = 0.2 \times 10^{-5} \times \frac{1}{760} = 2.635 \times 10^{-9} \text{ atm}$$

$$\Delta G^\circ = -4.575T \log (2.635 \times 10^{-9})^{1/2} \\ = 19.6T$$

$$40500 + 3.92T \log T - 29.5T = 19.6T$$

$$40500 - 49.1T_d + 3.92T_d \log T_d = 0$$

T	500	1000
Eq^n	21240	3160

by graphically $\rightarrow$

The lowest temperature = 1085°K

5.(b) Calculate the equilibrium constant and equilibrium partial pressure of oxygen for the reaction $\text{Zr} + \text{O}_2 = \text{Zr O}_2$ $\Delta G^\circ = 259940 + 2.33T \log T$

$- 59.12T$

at 2000K. Also predict the possibility of decomposing a pure zirconia crucible under a vacuum of 10^{-5}mmHg .



$$\Delta G^\circ = 259940 + 2.33 \times 2000 \log 2000 - 59.12 \times 2000$$

$$- RT \ln K = 156885$$

$$\ln K = \frac{156885}{-1.987 \times 2000} = -39.47$$

$$K = 7.217 \times 10^{-18}$$

$$K = \frac{a_{\text{ZrO}_2}}{a_{\text{Zr}} \cdot P_{\text{O}_2}} = \frac{1}{P_{\text{O}_2}}$$

$$P_{\text{O}_2} = \frac{1}{K} = 1.3858 \times 10^{17} \text{ mmHg}$$

$$Q = \frac{1}{P_{\text{O}_2}} = 10^5$$

Q) $K \therefore \text{Reaction Right to left}$

Zirconia crucible will be decomposed.

6.(a) In separate experiments, pure nitrogen at atmospheric pressure is passed over either mercury or a mercury-thallium amalgam (68.4 atom percent thallium). At temperature less than 100°C the vapor pressure of thallium is negligible, relative to that of mercury. The flow rate of nitrogen is controlled such that the gas is just saturated with mercury vapor as it passes over the pure mercury or the amalgam. The vapor carried over in the gas stream is condensed in a cold trapped and weighted. In experiments with the samples at 26°C , the weight of mercury trapped from the pure mercury and the amalgam were respectively 2.20 and 0.75 mg per hundred liter of nitrogen, the nitrogen volume being measured at 20°C and 1 atmosphere pressure. Assuming that equilibrium is continuously maintained over the samples, calculate the partial pressure of mercury over the amalgam and the activity coefficient of mercury in the amalgam at 26°C . (atomic weight of Hg = 200.6 g / mole)

Ans: $Hg_{(\text{pure}, l)} = Hg_{(\text{g}, \text{in } N_2)}$

$$K_P = \left(\frac{P_{\text{Hg}}}{a_{\text{Hg}}} \right)_{\text{pure Hg}} \dots\dots\dots (1)$$

$$= P_{\text{Hg}}(\text{pure Hg})$$

$Hg_{(l, \text{amalgam})} = Hg_{(\text{g}, \text{in } N_2)}$

$$K_P = \left(\frac{P_{\text{Hg}}}{a_{\text{Hg}}} \right)_{\text{amalgam}} \dots\dots\dots (2)$$

Assuming Hg vapor behave ideally, (1) = (2)

$$P_{Hg(pureHg)} = \left(\frac{P_{Hg}}{a_{Hg}} \right)_{\text{amalgam}}$$

$$A_{Hg} = \frac{P_{Hg(\text{amalgam})}}{P_{Hg(\text{pureHg})}} \dots\dots\dots (3)$$

$$\text{Volume of (100liter at } 20^{\circ}\text{C) at } 26^{\circ}\text{C} = 100 \times \frac{299}{273} = 102 \text{ liter}$$

$$M_{Hg} = 200.6 \text{ g / mole, } 2.2 \text{ mgHg occupies } \left(\frac{22.4}{200.6} \times 2.2 \right) = 0.239 \text{ ccat NTP.}$$

$$= 0.239 \times \frac{299}{273} \text{ ccat } 26^{\circ}\text{C}$$

$$= 0.262 \text{ cc at } 26^{\circ}\text{C}$$

$$\therefore \text{partial pressure of Hg over pure Hg} = \frac{0.262}{102000} \times 760$$

$$P_{Hg}^0 = 19.5 \times 10^{-4} \text{ mmHg}$$

$$\text{at NTP } 0.75 \text{ mg Hg occupies} = \frac{22.4}{200.6} \times 0.75 = 0.082 \text{ cc}$$

$$\text{at } 26^{\circ}\text{C } 0.75 \text{ mg Hg occupies} = 0.082 \times \frac{299}{273} = 0.089 \text{ cc}$$

$$\therefore \text{partial pressure of Hg over amalgam} = \frac{0.089}{102000} \times 760$$

$$P_{Hg} = 6.65 \times 10^{-4} \text{ mmHg}$$

$$(3) \quad a_{Hg} = \frac{P_{Hg}}{P_{Hg}^0} = \frac{6.65 \times 10^{-4}}{19.5 \times 10^{-4}} = 0.34$$

$$\rho_{Hg} = \frac{a_{Hg}}{N_{Hg}} = \frac{0.34}{0.316} = 1.08$$

6.(b) Chromium plates are bright annealed at 1000K in a wet hydrogen atmosphere. The pressure of wet hydrogen is 1atm. (i) Calculate the permissible water content in the hydrogen if there is to be no oxidation at 1000K (ii) Will annealed chromium plates be oxidized when cooled to 500K in the furnace atmosphere, as calculated in (i)? neglect the possibility of dissolution of hydrogen in chromium.

$$\text{Given: } 2\text{Cr} + 3(\text{H}_2\text{O}) = \text{Cr}_2\text{O}_3 + 3(\text{H}_2) \quad \Delta G^{\circ} = -91050 + 22.8T \text{ cal.}$$

Ans: $2\text{Cr} + 3(\text{H}_2\text{O}) = \text{Cr}_2\text{O}_3 + 3(\text{H}_2)$ $\Delta G^\circ = -91050 + 22.8T \text{ cal}$

$$\Delta G_T^\circ = -91050 + 22.8T$$

$$\Delta G_{1000}^\circ = -91050 + 22.8 \times 1000$$

$$-RT \ln K = -68250$$

$$\ln K = 34.348$$

$$K_{1000} = \left(\frac{P_{\text{H}_2}}{P_{\text{H}_2\text{O}}} \right)^3 = 8.265 \times 10^{14}$$

$$\frac{P_{\text{H}_2}}{P_{\text{H}_2\text{O}}} = 93836.27$$

$$\frac{1 - P_{\text{H}_2\text{O}}}{P_{\text{H}_2\text{O}}} = 93836.27$$

$$P_{\text{H}_2\text{O}} = 1.06 \times 10^{-5}$$

$$\Delta G_{500}^\circ = -91050 + 22.8 \times 500$$

$$-RT \ln K = -9650$$

$$\ln K = \frac{9650}{1.987 \times 500}$$

$$Q = K_{500} = 16533$$

$K > Q$. Left to Right

Cr. Plate will be oxidized at 500 °K